

Traffic & pedestrian microsimulation

What this tool produces · *Dynamic, animated digital models predicting exact congestion levels, travel delays, and collision risks resulting from a proposed change to the road layout or pedestrian network.*

G2 · Proof of Fit

Dimensions

Spatial

Environmental

Social / User

Preparing the workshop · *Traffic & pedestrian microsimulation*

G2 · Proof of Fit

Why this tool matters

Preventing the street from becoming a dangerous laboratory; using mathematics to ensure that every new bike lane or pedestrian crossing protects human life before the first line of paint is drawn.

Evidence we produce

Dynamic, animated digital models predicting exact congestion levels, travel delays, and collision risks resulting from a proposed change to the road layout or pedestrian network.

Duration & Material

Duration · 4 to 8 weeks for network coding, calibration, and scenario testing

Material · Microsimulation software (SUMO, VISSIM, CUBE), geometric road plans, real-time traffic data

Reference case

Transportation planners use SUMO (Simulation of Urban Mobility) and VISSIM to model complex intermodal traffic systems. Because these tools can simulate autonomous software agents (individual drivers, buses, and pedestrians interacting), planners can evaluate the precise impact of new transport policies, transit routes, or smart city technologies on overall network performance prior to physical deployment.

Suggested agenda · Brief (10 min) > Method walk-through (15 min) > Animation canvas (60-90 min) > Harvest (20 min)

Microsimulation parameters · agents · scenarios · KPIs

Duration
4 to 8 weeks for network coding, calibration,
and scena...

Agents

Pedestrians · cyclists · cars · transit · freight

1

2

3

4

5

6

7

Network

Geometry · signals · priorities · speeds

1

2

3

4

5

6

7

Scenarios

Today · planned design · stress tests

1

2

3

4

5

6

7

KPIs

Delay · queue · level-of-service · safety conflicts

1

2

3

4

5

6

7

Harvesting our *learnings*

After running the canvas, capture three layers of insight.

What inspires me

Stand-out signals, surprising stories, ideas to remember

How to apply it here

Concrete adaptations to our context, our team, our constraints

What we need to be autonomous

Skills, allies, resources, decisions still to secure

Each participant fills a copy. Then debrief together to surface patterns and next steps.